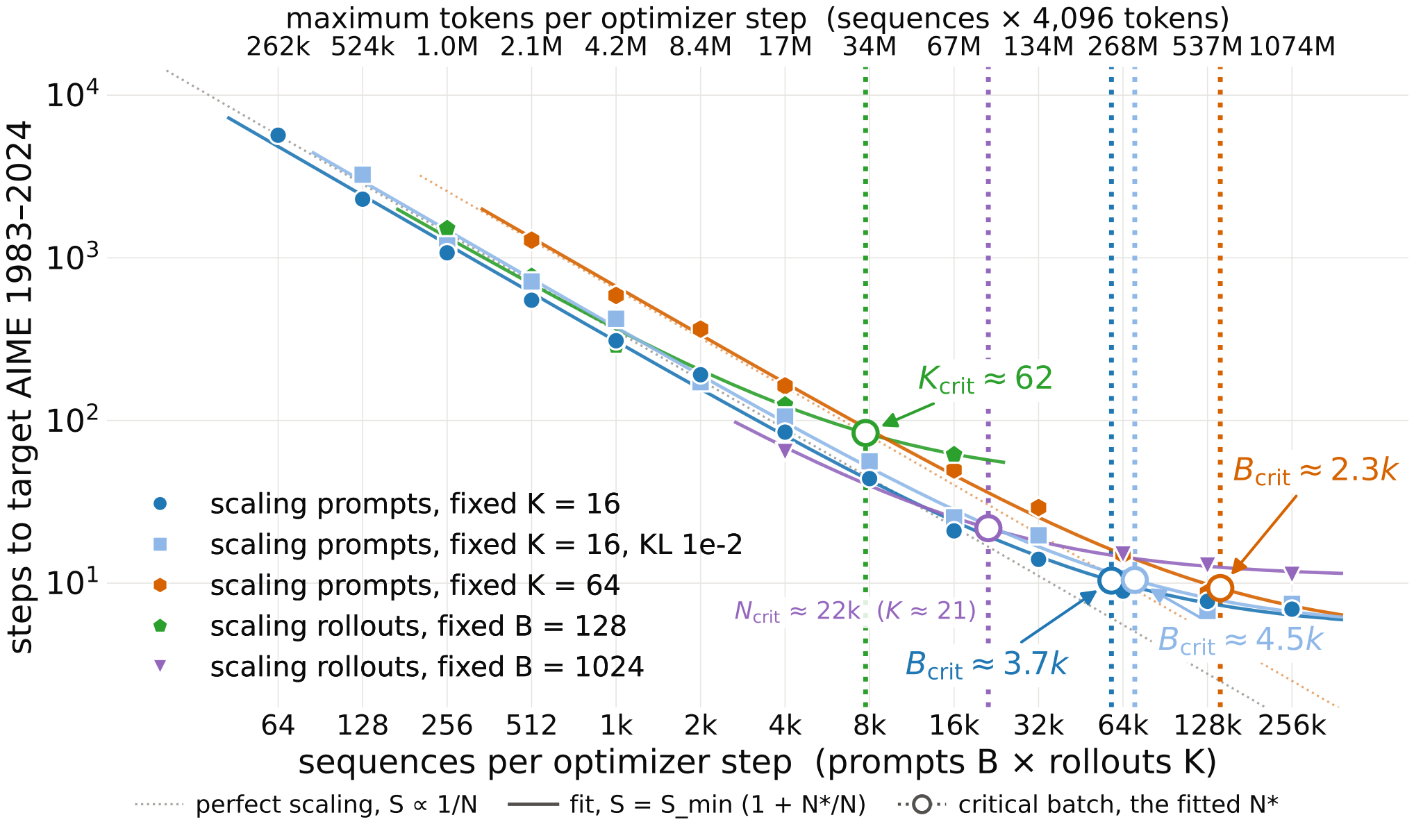
maximum tokens per optimizer step (sequences × 4,096 tokens)
262k 524k 1.0M 2.1M 4.2M 8.4M 17M 34M 67M 134M 268M 537M 1074M
steps to target AIME 1983–2024
10^4
10^3
10^2
10^1
scaling prompts, fixed K = 16
scaling prompts, fixed K = 16, KL 1e-2
scaling prompts, fixed K = 64
scaling rollouts, fixed B = 128
scaling rollouts, fixed B = 1024
$K_{crit} \approx 62$
$B_{crit} \approx 2.3k$
$N_{crit} \approx 22k$ ($K \approx 21$)
$B_{crit} \approx 4.5k$
$B_{crit} \approx 3.7k$
64 128 256 512 1k 2k 4k 8k 16k 32k 64k 128k 256k
sequences per optimizer step (prompts B × rollouts K)
perfect scaling, S ∝ 1/N
fit, S = S_min (1 + N*/N)
critical batch, the fitted N*